

# AADK 3 - Task 4 — Accessibility, Performance & Review

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## 1 Accessibility & Performance Evaluation

### ✓ A. Text Readability

- **Title (24 sp, Bold)** → Good visibility on most screens.
- **Body Text (16 sp)** → Acceptable, but may feel small on larger devices.
- **Contrast** → Must ensure strong contrast (e.g., dark text on light background).
- **Suggested contrast ratio:** At least 4.5:1 for accessibility compliance.

✓ Improvement Idea: Increase body text to **18 sp** for better readability.

### ✓ B. Image Resource

- **Planned:** birthday\_banner.png in drawable-nodpi.
- **Risk:** If image resolution is too large (e.g., 4000×3000 px), it increases memory usage.
- **Accessibility Concern:**
  - If image is decorative → `contentDescription = null`
  - If meaningful → Add proper description (e.g., “Birthday cake with balloons”).

✓ Improvement Idea:

- **Resize image** to approx. **1080 × 720 px**
- **Compress properly** (medium size < 400 KB)

### ✓ C. Layout Responsiveness

**Small Screens:**

- **Image height (200 dp)** might consume too much space.

- Text may feel crowded.

### Large Screens / Tablets:

- Content may look too centered and narrow.
- Better use spacing or max width constraints.

### ✓ Alternative Layout Idea:

- On larger screens → Use Row
  - Image on left
  - Text content on right
- Use Modifier.fillMaxWidth() with padding constraints.

### ✓ D. Touch Areas & Buttons

- Button must follow Android minimum touch target: **48 dp height**
- Ensure:
  - At least 8–16 dp spacing around button.
  - Adequate width for easy tapping.
- Button text should be clearly readable (14–16 sp minimum).

### ✓ Improvement Idea:

- Set button height explicitly to minHeight = 48 dp.

### 2 Review Table

| Element      | Potential Issue                                       | Proposed Fix                        | Priority |
|--------------|-------------------------------------------------------|-------------------------------------|----------|
| Banner Image | Large raw image (4000×3000 px) may affect performance | Resize to 1080×720 px, compress PNG | High     |
| Body Text    | 16 sp may be small for accessibility                  | Increase to 18 sp                   | Medium   |

| Element             | Potential Issue                | Proposed Fix                                            | Priority |
|---------------------|--------------------------------|---------------------------------------------------------|----------|
| Image Accessibility | No content description defined | Add meaningful contentDescription or null if decorative | High     |
| Button Size         | May not meet 48 dp minimum     | Enforce minimum height 48 dp                            | High     |
| Tablet Layout       | Content too centered/narrow    | Use Row layout for large screens                        | Medium   |

### 3 Mini Action Plan

#### ◆ Before Coding

1. Resize and compress banner image.
2. Finalize typography (Title 24 sp, Body 18 sp).
3. Decide whether image is decorative or meaningful (define contentDescription).

#### ◆ During Coding

1. Apply proper padding (16 dp screen margin).
2. Enforce 48 dp minimum button height.
3. Test layout using preview on different screen sizes.

#### ◆ After Testing

1. Run accessibility scanner (check contrast and touch targets).
2. Test on emulator (small phone + tablet).
3. Optimize if layout breaks or spacing feels inconsistent.

### 4 Reflection

Planning for accessibility and performance early prevents major redesign later. It ensures the UI is usable by all users, including those with visual or motor limitations. Optimized images and proper layout structure improve app speed and responsiveness. Overall, early review leads to a more professional, efficient, and inclusive final product.